

Physics 151 Syllabus – Fall 2026

General information

Instructor: Prof. Jiang-Xiazi Lin (jiangxiazi.lin@emory.edu)

Lab supervisor: Cory Donofrio (cjdonof@emory.edu)

PhysMentors: Ava Tholen (ava.tholen@emory.edu), Andrew Duh (andrew.duh@emory.edu), and Aishani Valluru (aishani.valluru@emory.edu)

Time and locations:

Lectures	Mon Wed 1:00pm - 2:15pm. Math & Science Center E208
Office hours	Tue 4-5pm and Thu 12-1pm. Math & Science Center N246
Mentor sessions	Mon 6-7pm at W201 (Andrew), Tue 6-7pm at E208 (Ava), Wed 6-7pm at W201 (Aishani)
Lab	See a separate Canvas page

Textbook:

Physics for Scientists and Engineers with Modern Physics, by Serway and Jewett, 10th edition

Course website:

- Canvas: all course information, including announcements and lecture notes (lecture and lab have separate Canvas sites)
- WebAssign: for homework (lecture and lab share the same WebAssign site)

Learning objectives

This is an introductory physics course presenting the foundations of classical mechanics to scientists and engineers. The course will cover topics related to mechanical motions, such as velocity, acceleration, momentum, force, energy, fluid mechanics, and waves. Some fundamental concepts of thermodynamics will also be included towards the end. Students will learn how to apply these abstract physics models both conceptually and mathematically to solve real-world problems and will see the role of physics in other disciplines. This course will include both lectures and lab work.

Prerequisites

Before taking this class, students should have learned math topics such as trigonometry and calculus, including the integral and derivative of polynomials and of trigonometric functions.

Physics knowledge is not a prerequisite.

Course materials

Textbook: Physics for Scientists and Engineers with Modern Physics, by Serway & Jewett, 10th edition, with Enhanced WebAssign (for homework)

Other editions of the text are usable as well. If you buy a paper copy from Emory's bookstore, it'll likely come pre-packaged with an I-have-already-paid-for-WebAssign access code. Alternatively, if you do not need a paper copy, you can pay WebAssign directly which will include access to an electronic version of our textbook.

Lab manuals will be freely distributed online.

Lecture notes will be posted on Canvas after each class on the same day.

Course requirements and grading

Homework (15% of final grade) is an essential component of this class. You must diligently work on the homework assignments to master the concepts and, consequently, perform well on the exams.

- Homework will be **published after each Wednesday lecture and be due at 1 pm the next Wednesday**. For special cases the deadline can be extended, but there are limitations: 1) Request must be emailed to me before the original deadline, 2) Each student has two chances for the semester.
- Homework will be completed online at the WebAssign website (<http://www.webassign.net>) which is associated with the textbook we use. You will need to either purchase the **access code** on WebAssign which includes an electronic textbook, or use the access code bundled with your textbook if you purchased your textbook from the Emory Bookstore. Access is free until 09/09/2026.
- You will need to enter our **Class Key: emory 6650 3919**. You only need to enter the key once. After that, you can just log in with your username and password.
- Group discussion and artificial intelligence (AI) usage on homework is not prohibited. After all, you must genuinely spend the effort to work on the problems and form the understandings by yourself. Variations of homework problems will appear in the midterm and final exams, so please make sure you understand how to approach each type of problems or models, and not just memorizing the superficial final answer.

Lab (20% of final grade) will be managed by Cory. You must pass both Lab AND Lecture to pass PHYS151 as a whole.

Mid-term exams (35% of final grade) There will be two mid-term exams. They will be held in class and in written form. The lower midterm score will account for 15% of the total grade and the higher midterm score will account for 20% of the grade.

Final exam (30% of final grade) will be held in person and in written form at the end of the semester. Time and date are determined by the College.

- For the mid-term or final exams, in the case of a conflict, please contact me as soon as possible, at least one week ahead of time. After-the-fact excuses for exam absence will require documentation and may not be acceptable. Exams will be based on lecture material, homework assignments, and questions and problems discussed in the lectures.

Course Grades 93.0 – 100 will be an A, 90.0 – 92.9 an A-, 87.0 – 89.9 a B+, 83.0 – 86.9 a B, 80.0 – 82.9 a B-, 77.0 – 79.9 a C+, 73.0 – 76.9 a C, 70.0 – 72.9 a C-, 67.0 – 69.9 a D+, 60.0 – 66.9 a D. A grade below 60.0 is failing. The Lab and the Lecture must each be passed independently for you to receive a passing grade in the course.

Office hours, PhysMentors sessions, and e-mail communications

Office hours are when you can directly talk to the professor. Just walk in during the time mentioned in the beginning of this syllabus. No prior appointment is needed. I strongly encourage you to take advantage of the office hours as the best setting to ask homework and other physics questions. During these office hours, I can use the blackboard to explain physical concepts with the clarity and conciseness that cannot be achieved via email.

PhysMentors Sessions are hosted by our physics mentors (also known as learning assistants) who are undergrad students who succeeded in this course before. You will have the opportunity to work in groups on physics problems and slowly discuss whatever physics issues you'd like. You are strongly encouraged to attend these sessions!

E-mail communications are efficient for short questions or urgent problems. Please use your emory.edu address to prevent being auto spammed. I will respond to emails approximately within 24 hours during the workdays. I might not be reachable on weekends and holidays. I highly value student feedback during the course. Please feel free to email me if you have any concerns or want to express your appreciation for how the course is being conducted. You can also drop a note in my mailbox in the Math & Science center, if you prefer anonymity.

Lab-related questions please be directed to Cory. Cory Donofrio's Office Hours can be found in the separate Canvas website that he administers for the Lab component of this course. Please consult him if you have questions about the weekly laboratory statistics lessons/homework that will be appearing on WebAssign. Cory is the person who writes and administers and handles the grades for those statistics materials.

Disability accommodations

It is the policy of Emory University to make reasonable accommodations for qualified students with disabilities. All students with special requests or need accommodation should make this request as soon as possible, accompanied by confirmation in writing (or electronic writing) from the Office of Accessibility Services. Any Office of Accessibility Services paperwork should be dated Fall 2025.

Course policy

Although collaboration is encouraged during the learning process, the homework submissions and the exams must be completed independently by each student. Cheating, lying, plagiarizing or any other act that violates the Emory Honor Code is strictly forbidden.

The purpose of this class is for everyone to learn physics. Please treat your classmates with respect both in and out of the classroom. It is inappropriate to judge your fellow students based on gender, race, your opinion of their physics ability, etc. If you have concerns about this sort of thing, feel free to contact me and/or the physics department chair: Prof. Connie Roth.

Course schedule

Date	Lecture number	Topic	Sections from <i>Serway and Jewett, 10th edition</i>
8/26	L1	Introduction & 1D motion part I	1.1; 1.4; 2.1-2.4
8/31	L2	1D Motion part II	2.5; 2.7-2.9
9/2	L3	Vectors and 2D motion part I	3.1-3.4; 4.1- 4.3
9/7	Labor Day – No Class		
9/9	L4	2D motion part II	4.4-4.6
9/14	L5	Newton's Laws part I	5.1-5.6
9/16	L6	Newton's Laws part II	5.7-5.8
9/21	L7	Newton's Laws part III	6.1-6.2
9/23	L8	Energy of a system part I	Ch. 7
9/28	L9	Energy of a system part II	Ch. 7 & 8
9/30	L10	Energy of a system part III	Ch. 7 & 8
10/5	L11	Review for Exam 1	Practice exam will be given
10/7	Exam 1		Ch. 2-8
10/12	Fall break – No class		
10/14	L12	Linear momentum part I	9.1-9.4
10/19	L13	Linear momentum part II	9.5-9.7
10/21	L14	Rotations part I	10.1-10.5, 10.7, 10.8

10/26	L15	Rotations part II	10.6, 10.9
10/28	L16	Rotations part III	11.1-11.4; 12.1; 12.3
11/2	L17	Gravitation	Ch. 13
11/4	L18	Fluid mechanics	14.1-14.6
11/9	L19	Review for Exam 2	Practice exam will be given
11/11	Exam 2		Ch. 9-14
11/16	L20	Oscillations	15.1-15.5
11/18	L21	Waves part I	16.1-16.3, 16.8-16.9
11/23	L22	Waves part II	17.1-17.4; 17.7
11/25	Thanksgiving break – No class		
11/30	L23	Thermodynamics part I	Ch. 18
12/2	L24	Thermodynamics part II	19.1-19.5
12/7	L25	Thermodynamics part III (maybe) and Review for Final Exam	(maybe Ch. 21), Practice exam will be given
12/9	L26	Example problem review	Selected problems
12/16	Final Exam (11:30 am - 2:00 pm)		All the topics